Reg. No. | | | | | | | | | | |

[22CSBSPH03]

FIRST SEMESTER B.Tech EXAMINATION, DECEMBER - 2022

FUNDAMENTALS OF PHYSICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

Physical constants

Charge of electron e=1.601 X 10^{-19} C,

Mass of electron m_e=9.1 X 10^{-31} Kg

Planck's constant h=6.625 X 10^{-34} J s,

Velocity of light $c=3\times10^{8}\ m/s$

Permittivity of free space $\varepsilon_0=8.85\times10^{-12}\ F/m$,

Permeability of free space $\mu_0=4\pi\times10^{-7}\ H/m$

Boltzmann's constant k_B=1.380649 × 10^{-23} JK^{-1}

PART-A (5×2=10)

ANSWER ALL QUESTIONS

1.	**a.**	List out the characterestics of Simple Harmonic Motion.	L-1	CO1	**02**
	b.	Describe double refraction.	L-2	CO2	**02**
	c.	Summarize the characterestics of matter waves.	L-2	CO4	**02**
	d.	Define population inversion and metastable state.	L-1	CO5	**02**
	e.	Explain reversible process in thermodynamics	L-2	CO6	**02**
		PART-B (6×5=30 Marks)			
2.	**a.**	What is force constant? Obtain an expression for force constant for springs which are parallely connected.	L-2	CO1	**05**
		OR			
	b.	Obtain the condition for resonance.	L-2	CO1	**05**
3.	**a.**	Distinguish interference and diffraction of light.	L-2	CO2	**05**
		OR			
	b.	What is coherence? Explain two types of coherence.	L-2	CO2	**05**
4.	**a.**	What are matter waves? How they are different from electromagnetic waves?	L-2	CO3	**05**
		OR			
	b.	An electron has a speed of 4.8x10^5 m/s accurate to 0.012%. With what accuracy can the position of electron be located.	L-2	CO3	**05**
5.	**a.**	What are Miller Indices? How do you find the Miller Indices of a given plane? Explain with an example.	L-2	CO4	**05**
		OR			
	b.	Explain the concept of effective mass of electrons and holes.	L-2	CO4	**05**
6.	**a.**	Discuss the point-to-point optical fiber communication system and mention its advantages over the conventional communication systems	L-2	CO5	**05**

		OR			
	b.	The numerical aperture of an optical fiber is 0.2 when surrounded by air. Determine the refractive index of its core if refractive index of cladding as 1.59. Also find the acceptance angle when it is in a medium of refractive index 1.33	L-2	CO5	**05**
7.	**a.**	Obtain an equation of continuity for current density.	L-2	CO6	**05**
		OR			
	b.	State and explain third law of thermodynamics and metion its importance interms of entropy.	L-2	CO6	**05**
		PART-C (6×10=60 Marks)			
8.	**a.**	Deduce an expression for amplitude and phase of a partical executing forced vibrations.	L-2	CO1	**10**
		OR			
	b.	Discuss the theory of damped oscillation with a neat diagram.	L-2	CO1	**10**
9.	**a.**	With the help of a neat diagram, describe Fraunhofer single slit experiment and hence obtain an expression for resultant amplitude.	L-2	CO2	**10**
		OR			
	b.	Calculate the wavelength of laser beam using the following data. Grating constant C = 5.08 x 10^{-5}m ; f = 80 cm Diffraction order (n) / Average distance (x in cm) 1 / 0.72 2 / 1.45 3 / 2.18 4 / 2.82 5 / 3.55	L-2	CO2	**10**
10.	**a.**	**i.** Deduce one dimensional time independent Schrodinger's wave equation for a particle. **ii.** An electron is bound in one dimensional potential well of width 0.3 nm. Find the energy values in eV in the second and third excited state.	L-2	CO3	**10**
		OR			
	b.	Assuming the time independent Schrodinger wave equation discuss the solutions for energy of a particle in a one dimensional potential well of infinite height and obtain the normalized wave function.	L-3	CO3	**10**
11.	**a.**	**i.** Explain different types of crystal structures with neat diagram. **ii.** Describe the lattice point, space lattice and unit cell.	L-2	CO4	**10**
		OR			

	b.	**i.** What is coordination number? Obtain the atomic packing factor for SC, BCC and FCC structures. **ii.** What is interplanar spacing? Deduce an expression for interplanar spacing in terms of Miller indices.	L-3	CO4	**10**
12.	**a.**	**i.** Discuss the various types of interaction of radiation with matter. **ii.** Deduce an expression for energy density in terms of Einstein's coefficients.	L-2	CO5	**10**
		OR			
	b.	**i.** Explain the construction and working of CO_2 LASER with energy level diagram. **ii.** The ratio of population of two energy levels out of which upper one corresponds to a metastable state is 1.059×10^{-30}. Find the wavelength of light emitted at 330K.	L-2	CO5	**10**
13.	**a.**	State Zeroth law of thermodynamics and describe the significance of zeroth law of thermodynamics. With the help of a neat diagram, explain Carnot's cycle.	L-3	CO6	**10**
		OR			
	b.	State and explain first law of thermodynamics. Discuss the application and its limitations.	L-2	CO6	**10**